

Week 3, September 25, 2024

NB: Due to conference travel, the session on Wednesday 18 is moved to a Friday afternoon in late October/November. More information soon. This means that the course week 3 will push one week forward.

Main topics

- Truncated SVD
- Tikhonov regularization

Material

From previous week: Kirsch 2-2.1 (you may skip 27-35). Appendix A, in particular Definition A.56, Theorem A.57, Theorem A.58. You can think of compact operators as integral operators. New material this week: Kirsch 2.2. In particular, pay attention to the equivalence of (2.15), (2.16) and (2.19).

Program

Tentative program:

13:00-13:30 Noise simulation, the Picard condition, TSVD and examples.

13:30-15:30 Exercise session - Truncated SVD

15:45-17:00 Tikhonov regularization

Exercises

You are highly encouraged to discuss and solve the exercises in small groups.

In the exercise session today we will among other things apply TSVD in order to solve inverse problems based on the model problems. We assume that you have finalized the numerical implementation of the Abel problem. For the other 2 problems, you can either finalize your own code or use the uploaded codes `HeatForward.m` and `BlurringForward.m`. This allows you to specify a discretization level n and matrix approximation A to be applied to a sampled vector x .

1. (a) Consider the differentiation problem

$$Kx(t) = \int_0^t x(s) \, ds.$$

Show that the following system

$$\begin{aligned}x_j(s) &= \sqrt{2} \cos \left(\left(j + \frac{1}{2} \right) \pi s \right) \\y_j(t) &= \sqrt{2} \sin \left(\left(j + \frac{1}{2} \right) \pi t \right) \\ \mu_j &= \frac{1}{\left(j + \frac{1}{2} \right) \pi}, \quad j = 1, 2, \dots\end{aligned}$$

satisfies $Kx_j = \mu_j y_j$.

(b) (For the mathematically inclined)

Characterize x_j by $\mu_j^2 x'' = x_j$ and $x_j(1) = 0$, $x_j'(0) = 0$.

Hint: x_j is an eigenfunction of K^*K .

2. For the heat problem, the forward operator K satisfies

$$K \sin(nx) = e^{-Tn^2} \sin(nx).$$

Conclude from this that the SVD of K is the singular system with

$$\phi_n(x) = \sqrt{\frac{2}{\pi}} \sin(nx), \quad \psi_n(x) = \sqrt{\frac{2}{\pi}} \sin(nx), \quad \mu_n = e^{-Tn^2}.$$

- Compute the SVD for the matrix A for the problem.
- Compare the singular vectors for A to the singular functions for K , and compare the singular values for A and K .
- Explain why the singular values for A level off.

3. For the Abel problem

$$Kx(t) = \int_0^t \frac{x(s)}{\sqrt{t-s}} ds$$

show that for $x(s) = 1$, $Kx(t) = 2\sqrt{t}$.

4. Consider the matrix A for Abel model problem.

- Sample the function

$$f(t) = \begin{cases} \sin(t)e^{-t}, & 0.1 < t < 0.9 \\ 0, & \text{otherwise,} \end{cases}$$

on the t -grid $h/2 : h : (1 - h/2)$ to obtain a vector x . Compute $y = Ax$.

- Compute the SVD of A and inspect the singular vectors and the decay of the singular values. Compare the decay to the decay $j^{-1/2}$. Is the Abel problem better posed than the numerical differentiation?

- Simulate Gaussian random noise

```
d = randn(size(y))
```

and scale it to have the predefined relative noise level .05 :

```
d = d/norm(d)*norm(y)*0.05
```

```
y_d = y + d
```

Compute the naive solution to $Ax = y^\delta$ (in Matlab by backslash). Plot the result, experiment with the discretization level and the noise level $\delta = 1\%$, 5% and 10% .

- Compute the TSVD solution to $Ax \approx y^\delta$.

5. Revisit the heat problem. Generate as above A, x, y, y^δ for the function

$$f(x) = x \sin(\pi - x) e^{-2x},$$

and relative noiselevel 1%.

Compute the matrix SVD and plot (in the same semilog plot) the singular values, the expansion coefficients (y^δ, y_j) and the reconstruction coefficients $\frac{(y^\delta, y_j)}{\mu_j}$. This plot is the so-called Picard plot. Experiment with other levels of noise and other functions f .

6. Fix $y \in L^2(0, \pi)$. Construct a function $d \in L^2(0, \pi)$ by defining specific Fourier coefficients so that $\|d\| < \delta \|y\|$ such that for $y^\delta = y + d$

$$\begin{aligned} \lim_{n \rightarrow \infty} (y^\delta, y_n) &= 0, \\ \lim_{n \rightarrow \infty} \frac{(y^\delta, y_n)}{\mu_n} &= \infty. \end{aligned}$$

7. For a given noiselevel δ , take the truncation $\alpha(\delta)$ in TSVD such that

$$\lim_{\delta \rightarrow 0} \alpha(\delta) = 0, \quad \lim_{\delta \rightarrow 0} \frac{\delta^2}{\alpha(\delta)} = 0.$$

Show that $R_\alpha(\delta)$ is a regularization strategy with admissible parameter choice rule $\alpha(\delta)$.

8. Compare the SVD analysis of a matrix and a compact operator in $L^2(a, b)$; what similarities and differences do you find?